MIL-C-3098/102A (USAF)
15 March 1975
SUPERSEDING
MIL-C-3098/102 (USAF)
10 July 1968

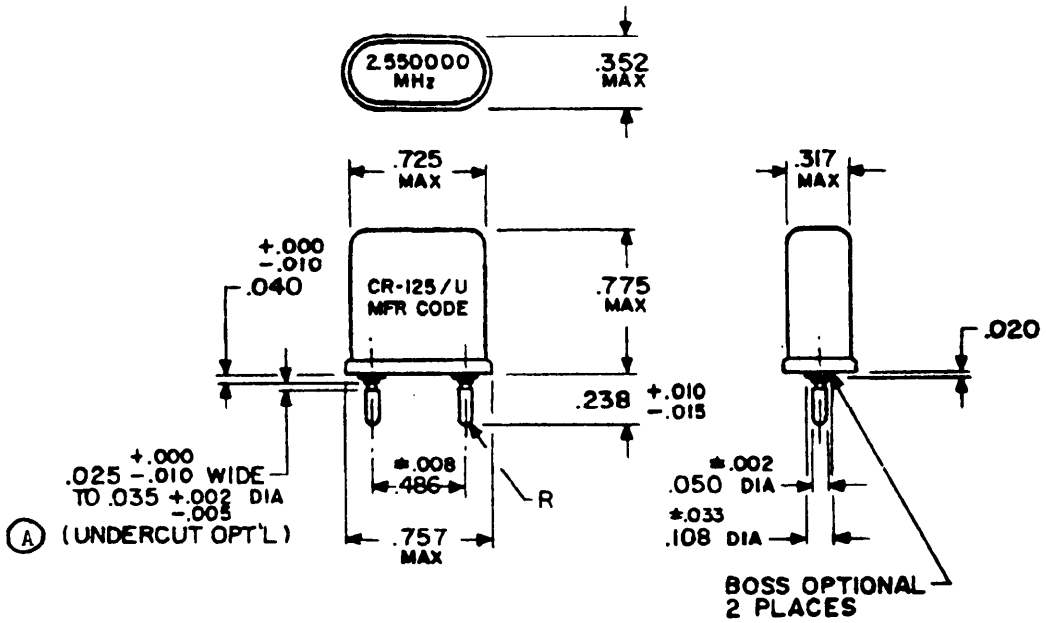
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-125/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the crystal unit described herein shall consist of this document and the latest issue of Military Specification MIL-C-3098.

Pertinent characteristics: 1.85 to 3.05 MHz; fundamental mode; noncontrolled; antiresonance.




| INCHES | MM | INCHES | MM | INCHES | MM | INCHES | MM |
|---|---|---|---|---|---|---|---|
| .002 | .05 | .020 | .51 | .050 | 1.27 | .486 | 12.34 |
| .005 | .13 | .025 | .64 | .108 | 2.74 | .725 | 18.42 |
| .008 | .20 | .033 | .84 | .238 | 6.05 | .757 | 19.23 |
| .010 | .25 | .035 | .89 | .317 | 8.05 | .775 | 19.69 |
| .015 | .38 | .040 | 1.02 | .352 | 8.94 | | |

NOTES:
1. DIMENSIONS ARE IN INCHES.
2. METRIC EQUIVALENTS ARE GIVEN FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH = 25.4 mm TO THE NEAREST .01 mm.

(A) DENOTES CHANGES

FSC 5955



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TABLE I. Requirements

| Requirement | Specified Value or Condition |
|---|---|
| Holder | HC-6/U (MIL-H-10056/2) |
| (A) Dimensions, marking, and configuration | See Figure |
| Frequency range | 1.85 to 3.05 MHz, inclusive |
| Capacitance, shunt | 7 pf, maximum |
| Frequency tolerance | ±0.005% |
| Mode of oscillation | Fundamental |
| Test set, rated drive level, and calibration values | See Table II |
| Antiresonant, load capacitance | 32.0 ±0.5 pf |
| Operating temperature range (noncontrolled) | -55° to +105°C, inclusive |
| Equivalent resistance | See Table III |
| Shock: | Below 2 MHz / 2 MHz and above |
| Frequency change permitted | ±0.001% / ±0.0005% |
| Equivalent resistance change permitted | ±15% or ±2 ohms (whichever is greater) |
| Vibration: 1/ | Below 2 MHz / 2 MHz and above |
| Frequency change permitted | ±0.001% / ±0.0005% |
| Equivalent resistance change permitted | ±15 % or ±2 ohms (whichever is greater) |
| Temperature cycling | ±0.001% |
| Aging: | |
| Frequency change permitted | 0.002% |
| Equivalent resistance | See Table III |
| Aging temperature | 105°C for 200 days |

1/ Method 204 of MIL-STD-202, test condition B, amplitude to produce 5 gravity units maximum.

TABLE II

DRIVE LEVEL CALIBRATION DATA

| Government Reference Standard Test Set | Band of Specified Frequencies Inclusive | Calibration Values | | | Rated Drive Level |
|---|---|---|---|---|---|
| | | Resistance | Crystal Current | Resistor Voltage Drop | |
| | MHz | Ohms | ma | Volt | mw |
| TS-330/TSM | 1.85 to 2.25 | 50 | 15 | - - | 10.0 ±2.0 |
| | 2.25+ to 3.05 | 40 | 15 | - - | 10.0 ±2.0 |



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TABLE III

EQUIVALENT RESISTANCE

| Frequency Range Inclusive | Maximum Resistance |
|---|---|
| MHz | Ohms |
| 1.85 to 1.999999 | 550 |
| 2.0 to 2.249999 | 500 |
| 2.25 to 2.999999 | 320 |
| 3.0 to 3.050000 | 175 |

NOTE: These crystal units were previously fabricated using Air Force Technical Order 12-1-20 and/or Collins Radio Company Specification 291-5969 thru 5970-00.

Custodians:
Air Force - 84

Review activities:
Air Force - 11, 17, 85

User activities:
Air Force - 19

Preparing activity:
Air Force - 84

(Project No. 5955-F171-10)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
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| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
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